

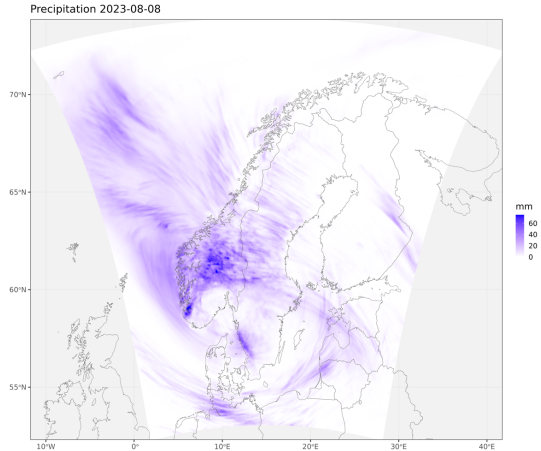
Impact of climate change on pluvial flooding damages in Norway

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Extreme precipitation event: Hans, 2023

- ▶ Extreme rainfall over Norway, August 2023
- ▶ 100+ local precipitation records
- ▶ Extreme-weather warnings: precipitation, flooding, landslides
- ▶ Significant damages on infrastructure and private property



Source: Norwegian Meteorological Institute (2023)

Contents

1. Motivation: Expert Committee on Climate Adaptation
2. Data used: Insurance claims, weather data, etc.
3. Model structure: Frequency and severity of claims
4. Damages in a future climate: Estimates and uncertainty

Expert Committee on Climate Adaptation (ECoCA)

- ▶ Appointed by the Norwegian government, July 2024

Mandate: Assess the socio-economic costs of climate change in Norway

- ▶ Concluding report presented to the Norwegian government, June 2026

Role for NR:

- ▶ Pluvial flooding large part of Norway's weather/climate-related damages
- ▶ Earlier experience w.r.t. impact of climate change on pluvial flooding¹
- ▶ Improve estimate of climate change impact on pluvial flooding damages

¹Heinrich-Mertsching, Wahl, Ordoñez et al. (2023): 'Assessing present and future risk of water damage using building attributes, meteorology, and topography'.

Pluvial water damages

- ▶ Caused by rain or snowmelt
- ▶ Water entering buildings from outside
- ▶ Excluded: river and lake flooding

Different mechanisms:

- ▶ Pluvial flooding: local rainfall, drainage, topography
- ▶ River-flooding: upstream flow, channel capacity, floodplain exposure

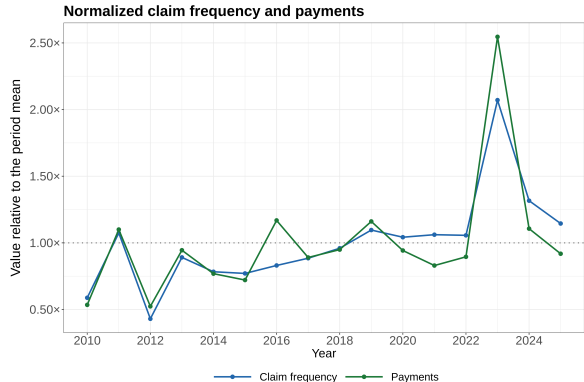


Photo: Bjørn Lytskjold / NVE

Sixteen years of building-level insurance data

Gjensidige data, 2010–2025:

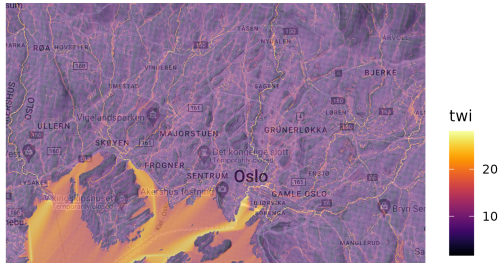
- ▶ Daily claim registration and building-level exposure
- ▶ Claims, payouts, and deductibles
- ▶ On the order of 200k contracts, and 40k claims
- ▶ Several residential and commercial building types



Source: Gjensidige building portfolio, 2010–2025

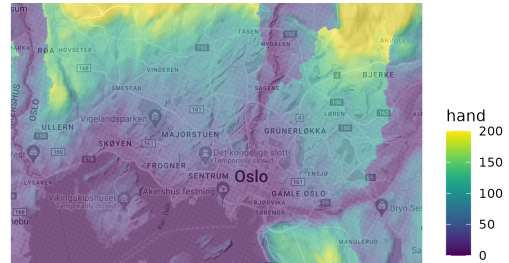
Topographic data sources

Topographic wetness index (TWI)



High values: drainage paths and flat areas²

Height above nearest drainage (HAND)



High values: elevated from drainage³

Land use: reduced to binary categories *urban* versus *non-urban*⁴

²Mattivi et al. (2019): 'TWI computation: a comparison of different open source GISs'.

³Nobre et al. (2011): 'Height Above the Nearest Drainage – a hydrologically relevant new terrain model'.

⁴Heggem, Mathisen and Frydenlund (2019): *AR50 - Arealressurskart i målestokk 1:50 000*.

Extreme precipitation measures

Daily precipitation

- ▶ Historical seNorge observations on a $1 \times 1 \text{ km}^2$ grid⁵
- ▶ Climate simulations from the Norwegian Centre for Climate Services⁶
- ▶ Values linked to each insured building by location

Two aggregation scales

- ▶ **Season-year:** daily values aggregated into indices measuring extreme precipitation
- ▶ **30-year reference:** local precipitation summaries describing the prevailing climate

Constructed 11 season-year indices, three 30-year reference measures.

⁵Lussana (2020): *seNorge observational gridded datasets. seNorge2018, version 20.05.*

⁶Dyrddal et al. (2025): *Klima i Norge – kunnskapsgrunnlag for klimatilpasning oppdatert i 2025.*

Static climate summaries to seasonal extremes

Retain building-level GAM framework:

Property, topographic, and regional effects similar to those in Heinrich-Mertsching et al.⁷ .

Previous model:

- ▶ Static climate summaries over a 30-year period
- ▶ Mean daily precipitation
- ▶ Mean temperature

This project: season-year extremes

1. *Absolute intensity*
Mean of the three wettest daily precipitation totals in each season-year
2. *Relative intensity*
Absolute intensity relative to the local 95th percentile of daily precipitation

⁷Heinrich-Mertsching, Wahl, Ordoñez et al. (2023): 'Assessing present and future risk of water damage using building attributes, meteorology, and topography'.

Total Damages Definition

Contract i , season-year Q (e.g. Winter-2023):

N_{iQ}
Number of claims

P_{iQ}^k
Payout for claim k

D_i
Deductible

Season-year Q damage for contract i :

$$S_{iQ} = \sum_{k=1}^{N_{iQ}} (D_i + P_{iQ}^k), \quad \sum_{k=1}^0 (D_i + P_{iQ}^k) = 0.$$

Total damages $T_{\mathcal{X}, \mathcal{T}}$: contracts collection $\mathcal{X}$, year-seasons collection $\mathcal{T}$:

$$T_{\mathcal{X}, \mathcal{T}} = \sum_{Q \in \mathcal{T}} \sum_{i \in \mathcal{X}} S_{iQ}.$$

Claim frequency model structure

N_{iQ} : number of claims on contract i in year-season Q ; l_{iQ} : insured days that season.

$$N_{iQ} \sim \text{NegBin}(\mu_{iQ}, \phi),$$

$$\log \left(\frac{\mu_{iQ}}{l_{iQ}} \right) = \beta_{\text{season}(Q)},$$

$$\begin{aligned} [\textit{property}] & \quad + \mathbf{x}_i^{\text{prop.}} \beta^{\text{prop.}} + \text{s}(\text{build year}_i), \\ [\textit{topography}] & \quad + \text{s}(\text{TWI}_i) + \text{s}(\text{HAND}_i) + \mathbf{x}_i^{\text{land use}} \beta^{\text{land use}}, \\ [\textit{region}] & \quad + \mathbf{x}_i^{\text{county}} \beta^{\text{county}} + \mathbf{x}_i^{\text{munic.}} \beta_{\text{land use}}^{\text{munic.}}, \\ [\textit{precipitation}] & \quad + \text{s}(\text{absolute precip}_{iQ}) + \text{s}(\text{relative precip}_{iQ}). \end{aligned}$$

Claim payment model structure

P_{iQ}^k : payout for claim k on contract i in year-season Q .

$$P_{iQ}^k \sim \text{Gamma}(\xi_{iQ}, \nu),$$

$$\log(\xi_{iQ}) = \beta_{\text{season}(Q)},$$

$$\begin{aligned} & \text{[property]} && + \mathbf{X}_i^{\text{prop.}} \beta^{\text{prop.}} + s(\text{build year}_i), \\ & \text{[topography]} && + s(\text{TWI}_i) + \mathbf{X}_i^{\text{land use}} \beta^{\text{land use}}, \\ & \text{[region]} && + \mathbf{X}_i^{\text{county}} \beta^{\text{county}} + \mathbf{X}_i^{\text{munic.}} \beta_{\text{land use}}^{\text{munic.}}, \\ & \text{[precipitation]} && + s(\text{absolute precip}_{iQ}) + s(\text{relative precip}_{iQ}). \end{aligned}$$

Inference framework: GAMs at scale with `mgcv`

$$g(E[Y]) = \beta_0 + \sum_j f_j(x_j) + X\beta$$

- ▶ Allows flexible non-linear effects of covariates⁸
- ▶ `bam()` designed for large GAMs⁹
- ▶ Smoothness well estimated from the data
- ▶ Enables use of millions of observations

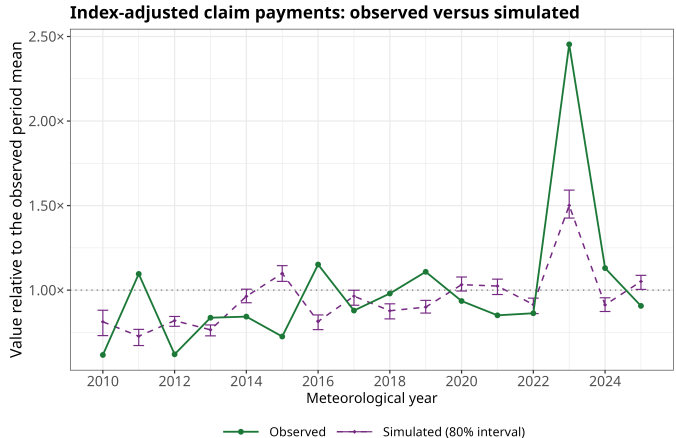
⁸S. Wood (2017): *Generalized Additive Models: An Introduction with R*.

⁹S. N. Wood et al. (2017): 'Generalized additive models for gigadata: modeling the U.K. Black Smoke Network daily data'.

Yearly cross-validation performance

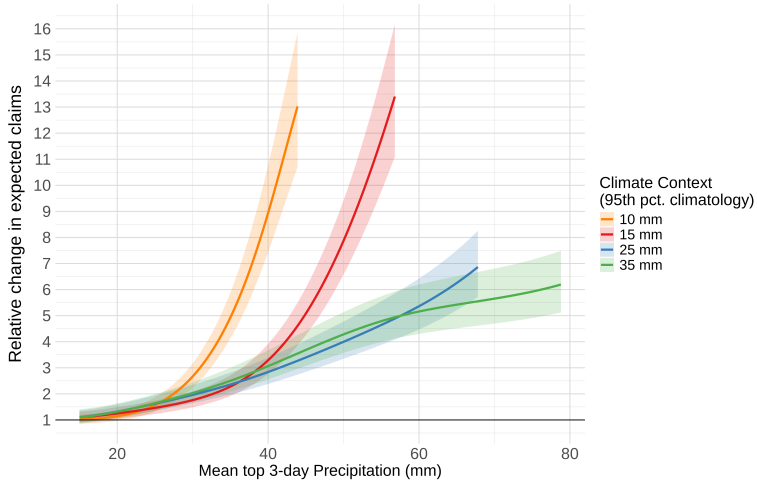
- ▶ Broadly well aligned
- ▶ Substantial yearly errors
- ▶ Underestimates uncertainty
- ▶ Full period, 50 draws: $-5.18 \pm 0.94\%$ error.

Intended use: long-period aggregates, not individual-year prediction.

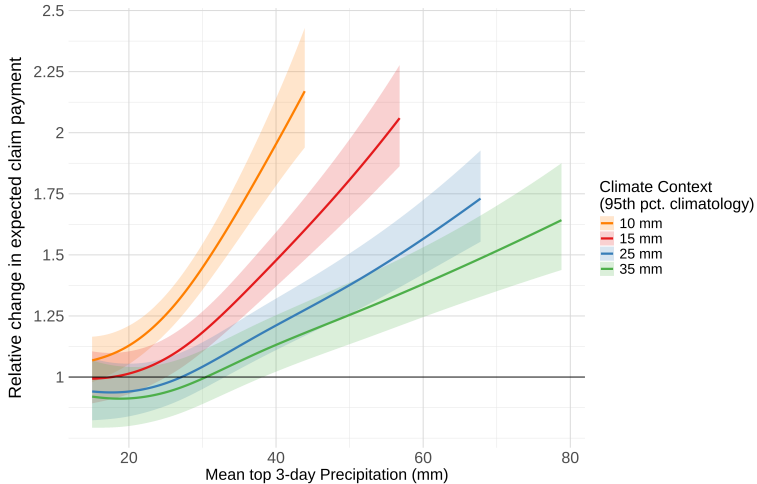


Source: Yearly cross-validation; annual payouts normalised by period mean

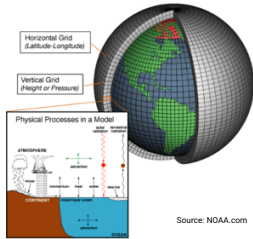
Combined precipitation effect - claim frequency



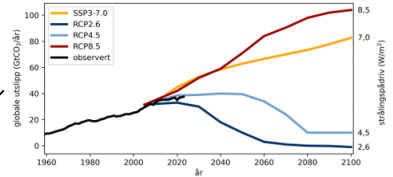
Combined precipitation effect - claim payment



Producing Climate Projections



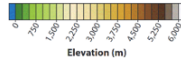
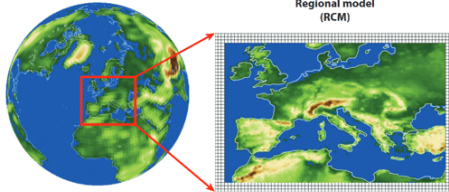
Global CO₂ emission scenarios



From the report *Klima i Norge* (2025)

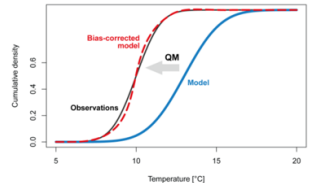
Global model
(AOGCM)

Regional model
(RCM)



From: F. Giorgi and W. Gutowski
(2015):
Regional Dynamical Downscaling
and the CORDEX initiative

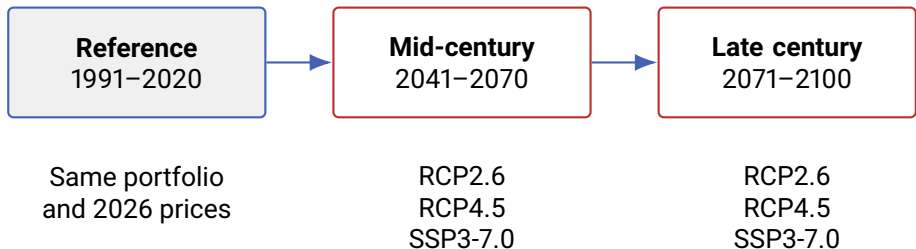
Bias correction



From: MeteoSwiss
tech rep no 270

Complex model
decisions,
uncertainty in
calibration data

Change the climate, hold building portfolio fixed



10 climate simulations × 2 bias adjustments × 10 posterior draws
= 200 relative cost changes per scenario, horizon, and geography

Source: Norwegian Climate Service Centre projections released in 2025

National change in total damages

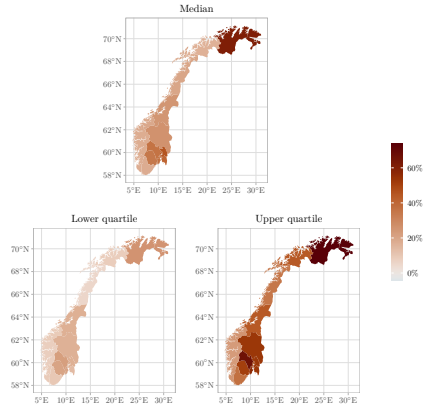
Future period	Emissions scenario	Lower quartile	Median	Upper quartile
2041–2070	Low	7.1%	9.5%	17.5%
	Intermediate	10.2%	14.9%	22.9%
	High	11.1%	18.3%	22.8%
2071–2100	Low	6.4%	9.9%	18.0%
	Intermediate	14.7%	23.4%	30.6%
	High	17.0%	33.2%	41.6%

Percent changes in total damages, across all of Norway.

Impact geographically uneven

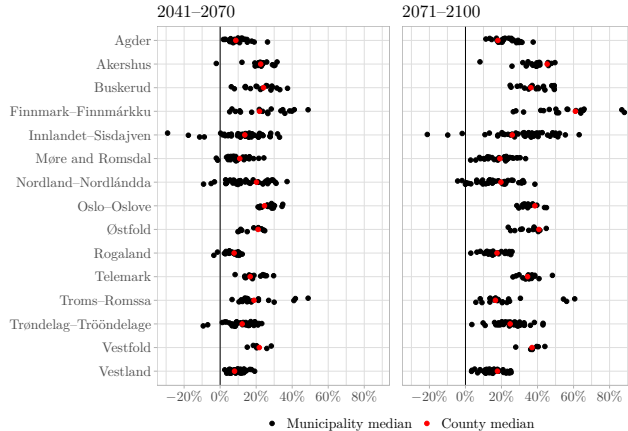
- ▶ High-emissions county medians:
 - ▶ 2041–2070: ca. +8% to +25%
 - ▶ 2071–2100: ca. +17% to +61%
- ▶ Largest increase in Finnmark, and southern mountain regions
- ▶ Lower increase in Western Norway

Change in total claim amount by county
From 1991–2020 to 2071–2100, under the high emissions scenario



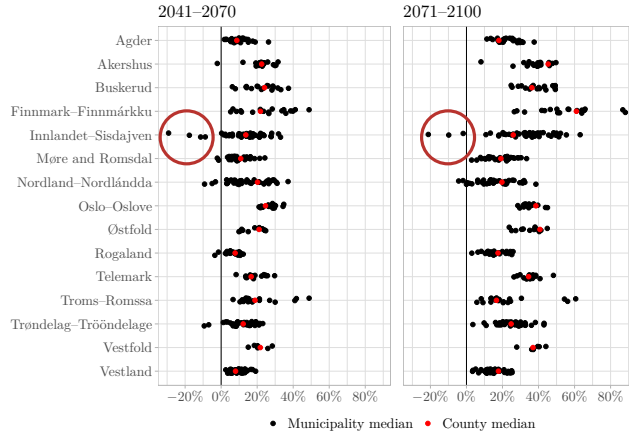
Geographical spread in median impact

Median change in total claim amount by municipality and county
Under the high emissions scenario, from 1991–2020 to ...

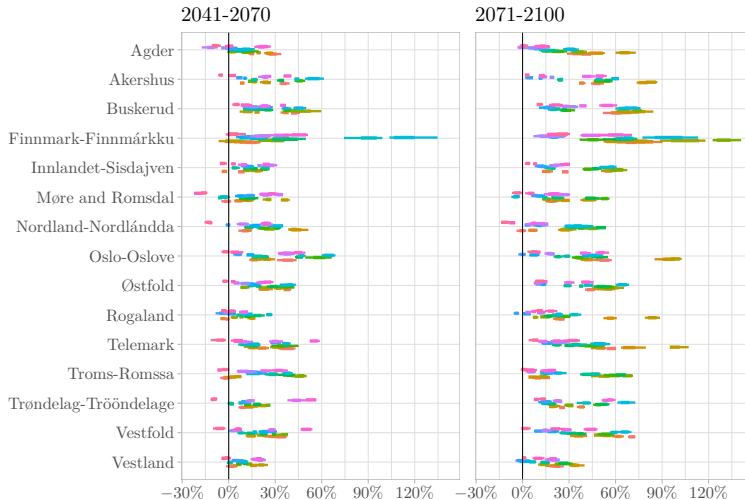


Geographical spread in median impact

Median change in total claim amount by municipality and county
Under the high emissions scenario, from 1991–2020 to ...



Change in total claim amount by county, for each of 20 climate projections Under the high emissions scenario, from 1991–2020 to ...



Summary

Conclusions

- ▶ High-emissions, 2071–2100: national median total damages increase by 33%
- ▶ Impact geographically uneven
- ▶ Greatest impact: Finnmark and southern mountain regions
- ▶ Substantial variation in impact from different climate simulations

Caveats

- ▶ Insurance data cover reported damages; claims below the deductible unobserved
- ▶ Ignore spatially correlated errors, clearly underestimating uncertainty
- ▶ Not representing short-term extreme precipitation events